Practice Exam 1

Sixty-five multiple-choice and multiple-response questions, timed to the length of the real examination.

QUESTIONS	TIME LIMIT	EXAM
65 12 multiple-response	90 minutes · 83 s per question	1 / 3 no question is repeated

B E F O R E Y O U S TA R T

1. Set a timer for 90 minutes and work without notes, documentation, or a browser.
2. Record every answer on the response sheet near the back. Do not write in the question pages if you plan to retake this exam.
3. Questions marked S E L E C T T WO require exactly two answers. There is no partial credit — both must be right.
4. Nothing is deducted for a wrong answer, so answer every question. Flag the ones you guessed and revisit them if time allows.
5. When the timer ends, stop, then score yourself with the answer key and read the explanation for every question you missed.

THIS EXAM COVERS

DOMAIN		Q	%
D1	Fundamentals of AI and ML	12	18
D2	Fundamentals of Generative AI	16	25
D3	Applications of Foundation Models	19	29
D4	Guidelines for Responsible AI	9	14
D5	Security, Compliance, and Governance	9	14
Total		65	100

PA C I N G C H E C K P O I N T S

Q17	Q33	Q49	Q65
after 23 min	after 45 min	after 68 min	at 90 min

11 questions cover the revised objectives: MCP, AgentCore, context engineering, distillation, Kiro, Strands, LLM-as-a-judge.

1 How does AWS Identity and Access Management (IAM) help ensure security and compliance?

A Data Backup Management
B Automated Data Classification
C Resource Provisioning and Scaling
D Granular Access Control

2 Which of the followings are best practices for creating effective prompts? S E L E C T T WO

A Be clear and concise
B Always use negative prompting
C Include relevant context
D Don't provide details so that the foundational modal is not limited
E Always use a high temperature

3 How does SageMaker Ground Truth support the capabilities needed for Reinforcement Learning from Human Feedback (RLHF)?

A SageMaker Ground Truth is specifically designed for real-time decision-making in autonomous systems, bypassing the need for any data labeling
B SageMaker Ground Truth uses pre-trained models to eliminate the need for human feedback in the reinforcement learning process
C SageMaker Ground Truth automatically generates synthetic data for training reinforcement learning models without any human intervention
D SageMaker Ground Truth enables the creation of high-quality labeled datasets by incorporating human feedback in the labeling process, which can be used to improve reinforcement learning models

4 What is a primary characteristic of diffusion models in generative AI?

A They use a fixed set of rules to generate outputs based on input patterns.
B They mimic human brain neural networks to create realistic outputs through backpropagation.
C They generate data by progressively adding noise to the input and then learning to reverse the noise process.
D They rely on labeled data to classify inputs into predefined categories.

5 Which metric primarily focuses on evaluating the similarity between a machine-generated summary and a human-written one by checking for matching sequences of words?
A BERTScore
B BLEU
C Precision
D ROUGE

6 Consider a scenario where a fully-managed AWS service needs to be used for automating the extraction of insights from legal briefs such as contracts and court records. What do you recommend?
A Amazon Transcribe
B Amazon Translate
C Amazon Rekognition
D Amazon Comprehend

7 A manufacturing company aims to leverage Amazon Bedrock to create a generative AI application that automates the monitoring of inventory levels, sales data, and supply chain information. The application should also recommend optimal reorder points and quantities to enhance operational efficiency. What do you recommend?
A Agents for Amazon Bedrock
B Watermark detection for Amazon Bedrock
C Guardrails for Amazon Bedrock
D Knowledge Bases for Amazon Bedrock

8 A telecom company wants to help its customer service agents provide better customer service by leveraging generative AI. Which of the following is the best fit for the given use case?
A Amazon Q in QuickSight
B Amazon Q Business
C Amazon Q Developer
D Amazon Q in Connect

9 Which Amazon SageMaker service will help you track the Machine Learning models that are hosted on endpoints for real-time inference ?

A Amazon SageMaker JumpStart
B Amazon SageMaker Model Dashboard
C Amazon SageMaker Clarify
D Amazon SageMaker Ground Truth

10 What is the key difference between prompt engineering and context engineering?

A Prompt engineering trains a model, while context engineering deploys the model
B Prompt engineering focuses on how instructions are written, while context engineering designs the system that assembles the relevant information shown to the model
C Prompt engineering applies only to text models, while context engineering applies only to image models
D Prompt engineering reduces model size, while context engineering increases model size

11 In the development of AI agents, which two factors are most critical for ensuring that the agent can effectively adapt to new and unpredictable environments? S E L E C T T WO

A Enabling the agent to receive real-time feedback and adjust its actions accordingly
B Pre-programming a comprehensive set of rules for every possible scenario
C Relying on a static dataset that contains all possible interactions the agent might encounter
D Limiting the agent's decision-making process to predefined outcomes for consistency
E Incorporating reinforcement learning to allow the agent to learn from its environment over time

12 How does Amazon Bedrock use a teacher model during a model-distillation job?

A It uses teacher responses to create synthetic training data for fine-tuning the student model
B It converts the teacher model into a vector database
C It uses the teacher only to encrypt the student's parameters
D It deploys the teacher and student behind the same real-time endpoint without customization

13 Which AWS services are specifically designed to aid in monitoring machine learning models and incorporating human review processes? S E L E C T T WO
A Amazon SageMaker Feature Store
B Amazon SageMaker Ground Truth
C Amazon SageMaker Data Wrangler
D Amazon Augmented AI (Amazon A2I)
E Amazon SageMaker Model Monitor

14 Which of the following best describes an action group in the context of setting up an AI agent?
A A sequence of user interactions
B A group of unrelated functions
C A collection of tasks that the agent can perform to help the user
D A set of predefined manual tasks

15 Which of the following are examples of unsupervised learning? S E L E C T T WO
A Clustering
B Neural network
C Sentiment analysis
D Decision tree
E Dimensionality reduction

16 A team wants to investigate an agent's production latency, errors, tool calls, and execution path. Which approach should it use?
A Enable AgentCore observability and use Amazon CloudWatch metrics, traces, and spans
B Increase the model's maximum output-token setting
C Fine-tune the model on its CloudTrail log files
D Replace all external tools with few-shot examples

17 What is a key difference between labeled data and unlabeled data in the context of machine learning?

A Labeled data is annotated with output labels that provide specific information about each data point and is used for supervised learning, whereas, unlabeled data lacks such annotations and is used for unsupervised learning
B Labeled data is inherently more complex to process due to the lack of structure, whereas unlabeled data is easier to handle because it has clear annotations
C Labeled data can only be used for unsupervised learning algorithms, whereas, unlabeled data is exclusively for supervised learning algorithms
D Labeled data consists of raw information with no tags or annotations, while unlabeled data includes tags that provide meaning to the data

18 What is a key advantage of using foundational models in generative AI applications?

A They can only be used for specific domains and require training from scratch for each new application
B They eliminate the need for reinforcement learning in AI systems
C They are pre-trained on vast datasets, allowing them to be fine-tuned for specific tasks with minimal additional training
D They are primarily designed to process structured data, such as spreadsheets and databases

19 Which AgentCore Memory feature most directly helps an agent personalize later conversations based on a preference stated in an earlier session?

A Short-term memory that is limited to a single model response
B Long-term memory that extracts and stores user preferences across sessions
C Gateway semantic tool search
D Runtime container isolation

20 A company is using the Amazon Titan Text model with Amazon Bedrock. In which of the following scenarios is the model most likely to hallucinate?

A When temperature is set to 0.5
B When temperature is set to 1
C When temperature is set to 0
D Temperature has no impact on hallucinations

21 Which of the following are the features of Amazon SageMaker JumpStart? S E L E C T T WO

A You can build highly accurate ML models using a visual interface without any code
B Your inference and training data will be used to train the base model
C SageMaker JumpStart provides only public models. Proprietary models are not supported by SageMaker JumpStart
D Pre-trained models are fully customizable for your use case with your data
E You can evaluate, compare, and select Foundation Models quickly based on pre-defined quality and responsibility metrics

22 How does Amazon Bedrock evaluate the pricing for image models?

A Based on the compute time used
B Based on the number of input tokens
C Based on the number of images generated
D Fixed monthly subscription fee

23 An insurance company is transitioning to AWS Cloud and wants to use Amazon Bedrock for product recommendations. The company wants to supplement organization-specific information to the underlying Foundation Model (FM). Which of the following represents the best-fit solution for the given use case?

A Fine-tune the base Foundation Model (FM) used by Amazon Bedrock by leveraging the contextual information from the company's private data
B Use Knowledge Bases for Amazon Bedrock to supplement contextual information from the company's private data to the FM using Reinforcement Learning from Human Feedback (RLHF)
C Use Knowledge Bases for Amazon Bedrock to supplement contextual information from the company's private data to the FM using Retrieval Augmented Generation (RAG)
D Implement Reinforcement Learning from Human Feedback (RLHF) in Amazon Bedrock by leveraging the contextual information from the company's private data

24 A company needs a solution that can convert text into human speech so that it can offer audio courses in multiple languages. Which AWS service is the best fit for this use case ?

A Amazon Lex
B Amazon Translate
C Amazon Polly
D Amazon Comprehend

25 What are the applications of large language models (LLMs)?

A LLMs are used for generating human-like text, translating languages, summarizing text, and answering questions based on large datasets

B LLMs are used for creating videos from textual descriptions

C LLMs are used for designing and generating 3D models for use in various applications such as gaming, virtual reality, or industrial design

D LLMs are used to synthesize realistic human speech from text inputs

26 What is the bias versus variance trade-off in machine learning?

A The bias versus variance trade-off is a technique used to improve model performance by increasing both bias and variance simultaneously to achieve better generalization

B The bias versus variance trade-off involves choosing between a model with high complexity that may capture more noise (high bias) and a simpler model that may generalize better but miss important patterns (high variance)

C The bias versus variance trade-off refers to the balance between underfitting and overfitting, where high bias leads to overfitting and high variance leads to underfitting

D The bias versus variance trade-off refers to the challenge of balancing the error due to the model's complexity (variance) and the error due to incorrect assumptions in the model (bias), where high bias can cause underfitting and high variance can cause overfitting

27 A team repeatedly explains its architecture, naming conventions, and preferred libraries in every Kiro conversation. Which capability should it configure?

A Steering files in .kiro/steering/

B A higher model temperature

C An Amazon Bedrock distillation dataset

D A CloudTrail trail containing source code

28 What is a key difference between Amazon Bedrock and Amazon SageMaker JumpStart?

- A Amazon SageMaker JumpStart is designed for building and scaling machine learning models, whereas Amazon Bedrock is used for real-time data analytics
- B Amazon Bedrock provides foundational models for generative AI applications, whereas Amazon SageMaker JumpStart offers pre-built solutions and one-click deployment for various machine learning models
- C Amazon SageMaker JumpStart provides foundational models for generative AI applications, whereas Amazon Bedrock offers pre-built solutions and one-click deployment for various machine learning models
- D Amazon Bedrock is designed for building and scaling machine learning models, whereas Amazon SageMaker JumpStart is used for real-time data analytics

29 In the context of the AWS Shared Responsibility Model, which statement best describes the security responsibilities of both AWS and the customer when using Amazon Bedrock for generative AI applications?

- A AWS handles all aspects of security for Amazon Bedrock, relieving the customer of any security responsibilities
- B The customer is responsible for the entire security stack, including the underlying infrastructure and the AI models
- C AWS is responsible for the security of the AI models and customer data, while the customer is responsible for securing the physical infrastructure
- D AWS is responsible for securing the infrastructure that runs Amazon Bedrock, while the customer is responsible for securing their data and managing access controls

30 How does the inference parameter Response length influence the model response for Amazon Bedrock?

- A Influences the percentage of most-likely candidates that the model considers for the next token
- B Specifies the sequences of characters that stop the model from generating further tokens
- C Influences the number of most-likely candidates that the model considers for the next token
- D Specifies the minimum or maximum number of tokens to return in the generated response.

31 A company needs to extract handwritten words and letters from scanned documents. Which ML-powered AWS service is the right fit for this requirement?

A Amazon Transcribe
B Amazon Textract
C Amazon Rekognition
D Amazon Kendra

32 Which of the following options summarize the differences between Amazon Q and Amazon Bedrock? S E L E C T T WO

A Amazon Q is a generative AI–powered assistant that allows you to create pre-packaged generative AI applications, whereas, Amazon Bedrock provides an environment to build and scale generative AI applications using a Foundation Model (FM)
B With Amazon Bedrock, you can choose the underlying Foundation Model. However, Amazon Q does not allow you to choose the underlying Foundation Model
C Both Amazon Q and Amazon Bedrock are generative AI-powered assistants that allow you to create pre-packaged generative AI applications
D Amazon Bedrock is a generative AI–powered assistant that allows you to create pre-packaged generative AI applications, whereas, Amazon Q provides an environment to build and scale generative AI applications using a Foundation Model (FM)
E With Amazon Q, you can choose the underlying Foundation Model. However, Amazon Bedrock does not allow you to choose the underlying Foundation Model

33 In the context of security and privacy for AI systems on AWS, what is the primary difference between threat detection and vulnerability management?

A Threat detection and vulnerability management both exclusively focus on compliance with regulatory requirements
B Threat detection involves real-time monitoring and identification of active threats, whereas vulnerability management is about identifying, assessing, and mitigating security weaknesses
C Threat detection is concerned with data encryption and access controls, while vulnerability management deals with incident response and recovery
D Threat detection focuses on identifying potential weaknesses in the system, while vulnerability management continuously monitors for malicious activities

34 Which of the following is correct regarding the training set, validation set, and test set used in the context of machine learning? S E L E C T T WO

A Test set is used for hyperparameter tuning
B Validation sets are optional
C Validation set is used to determine how well the model generalizes
D Test set is used to determine how well the model generalizes
E Test sets are optional

35 Which AWS service offers pre-trained and customizable computer vision (CV) capabilities?

A Amazon DeepRacer
B Amazon SageMaker
C Amazon Rekognition
D Amazon Textract

36 Which of the following are key advantages of using "few-shot" prompting in prompt engineering? S E L E C T T WO

A It completely eliminates randomness from the model's responses.
B It makes the model generate responses faster by limiting token usage.
C It helps the model understand the task by providing examples.
D It reduces the need for providing task-specific examples.
E It allows for more accurate results with complex tasks by guiding the model.

37 What sequence best represents the core Strands agent loop?

A Invoke the model, execute a requested tool, return the tool result to the model, and repeat until a final response
B Train a new foundation model, deploy it, and delete the training data after every user message
C Retrieve every tool result before the model sees the user's request
D Ask a human to manually select and execute every tool call

38 Which of the following is correct regarding data security and compliance aspects of Amazon Bedrock?

A Your data is only used to improve the base Foundation Models (FMs), however, it is not shared with any model providers
B Your data is not used to improve the base Foundation Models (FMs) and it is not shared with any model providers
C Your data is not used to improve the base Foundation Models (FMs), however, it is shared with the model providers for model optimization
D Your data is used to improve the base Foundation Models (FMs) and it is also shared with the model providers for model optimization

39 Which of the following scenarios best illustrates algorithmic bias?

A A human resources manager hires candidates based on personal interviews without considering their resumes
B A weather prediction model occasionally makes incorrect forecasts due to random fluctuations in weather patterns
C A hiring algorithm consistently prefers candidates from a particular gender, even though the candidates' qualifications are similar across genders
D A customer service representative resolves complaints based on their judgment rather than company policy

40 Which Kiro hook action is the better choice for running a fixed linter command whenever a relevant event occurs?

A Shell Command, because the action is deterministic
B Agent Prompt, because linting always requires creative reasoning
C Model distillation, because it converts source code into weights
D Long-term memory, because it executes operating-system commands

41 Which of the following generative AI techniques are used in Amazon Q Business web application workflow? S E L E C T T WO

A Large Language Model (LLM)
B Retrieval-Augmented Generation (RAG)
C Generative adversarial network (GAN)
D Variational autoencoders (VAE)
E Diffusion Model

42 Which feature of Amazon Macie enables organizations to automatically identify and safeguard sensitive data within their Amazon S3 buckets?
- A Access Analyzer
- B Data Lifecycle Manager
- C Automated Data Discovery
- D S3 Object Lock

43 What is the role of vector embeddings in Retrieval-Augmented Generation (RAG)?
- A They represent words in a vector space to understand semantics
- B They reduce the storage requirements for databases
- C They eliminate the need for user input
- D They replace traditional machine learning models

44 Which of the following is correct regarding the techniques used to improve the performance of a Foundation Model (FM)?
- A Neither Fine-tuning nor Retrieval-augmented generation (RAG) changes the weights of the FM
- B Fine-tuning does not change the weights of the FM whereas Retrieval-augmented generation (RAG) changes the weights of the FM
- C Fine-tuning changes the weights of the FM whereas Retrieval-augmented generation (RAG) does not change the weights of the FM
- D Both Fine-tuning and Retrieval-augmented generation (RAG) change the weights of the FM

45 Which AWS services/tools can be used to implement Responsible AI practices? S E L E C T T WO
- A AWS Audit Manager
- B Amazon SageMaker Model Monitor
- C Amazon Inspector
- D Amazon SageMaker Clarify
- E Amazon SageMaker JumpStart

46 Which AWS service/feature offers a choice of high-performing Foundation Models (FMs) and the ability to privately customize the FMs with your data?
A Amazon Q in QuickSight
B AWS Inferentia
C Amazon Q Developer
D Amazon Bedrock

47 Which of the following parameters can influence the creativity of a language model's output? S E L E C T T WO
A Temperature
B Input data
C Context length
D Top-p
E Model size

48 When configuring an AgentCore Gateway, what is the distinction between inbound and outbound authorization?
A Inbound authorization controls access to the gateway; outbound authorization controls how the gateway accesses target resources
B Inbound authorization encrypts training data; outbound authorization distills a teacher model
C Inbound authorization selects an LLM; outbound authorization selects an embedding model
D Inbound authorization applies only to prompts; outbound authorization applies only to responses

49 Which of the following best represents the capabilities of Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs)?
A Both RNNs and CNNs are used for video analysis
B While CNNs are used for single image analysis, RNNs are used for video analysis
C Both RNNs and CNNs are used for single image analysis
D While RNNs are used for single image analysis, CNNs are used for video analysis

50 A company wants to improve the performance of a Foundation Model (FM) being used in Amazon Bedrock. Which of the following lists the underlying techniques in the increasing order of complexity for implementing a solution?

A Retrieval Augmented Generation (RAG), Fine-tuning, Prompt engineering
B Prompt engineering, Retrieval Augmented Generation (RAG), Fine-tuning
C Prompt engineering, Fine-tuning, Retrieval Augmented Generation (RAG)
D Retrieval Augmented Generation (RAG), Prompt engineering, Fine-tuning

51 What is the difference between computer vision and image processing?

A Computer vision and image processing are identical fields with no distinct differences in their applications or techniques
B Computer vision focuses on enhancing and manipulating images for visual quality, whereas image processing involves interpreting and understanding the content of images to make decisions
C Image processing uses machine learning algorithms, while computer vision relies solely on pre-programmed rules
D Image processing focuses on enhancing and manipulating images for visual quality, whereas computer vision involves interpreting and understanding the content of images to make decisions

52 Which of the following aptly summarizes the capabilities of Foundation Models?

A Foundation models are limited to simple data processing tasks and cannot handle complex operations
B Foundation models are designed to work exclusively with structured data and cannot process unstructured data like text or images
C Foundation models can perform a wide range of tasks across different domains by leveraging their extensive pre-training on large datasets
D Foundation models can only perform a single task they were specifically trained for

53 Which Amazon SageMaker service/tool offers a centralized view of all the Machine Learning models created in your AWS account?

A Amazon SageMaker Model Dashboard
B Amazon SageMaker Model Monitor
C Amazon SageMaker Clarify
D Amazon SageMaker Ground Truth

54 The marketing department at a media company wants to leverage Amazon Bedrock for making creative scripts for an upcoming ad campaign. What do you recommend?

A Use lower Top-P to get more creative responses for the same prompt on Amazon Bedrock
B Use higher Top-P to get more creative responses for the same prompt on Amazon Bedrock
C Use lower Temperature to get more creative responses for the same prompt on Amazon Bedrock
D Use higher Temperature to get more creative responses for the same prompt on Amazon Bedrock

55 A developer is working on an AI application for predicting customer churn. The developer is collaborating with a research team to ensure the best model is selected for the application. The application needs to accurately identify customers who are likely to leave the service within the next six months. What should the developer ask the research team to do in order to ensure that the best model is selected for the AI application?

A Determine the cost constraints for model training
B Define the target audience of the application broadly
C Identify potential data sources for the application
D Define the use case of the application narrowly

56 Which practices improve the reliability of an LLM-as-a-judge evaluation? S E L E C T T WO

A Define clear evaluation criteria and rating scales
B Use representative evaluation examples and validate results against human judgment
C Assume the evaluator model cannot have bias
D Change the rubric for every response after viewing the answer
E Use only easy examples that all models answer correctly

57 What is the primary distinction between discriminative models and generative models in the context of generative AI?

A Discriminative models are used to generate new data, while generative models are used only for classification
B Generative models focus on generating new data from learned patterns, whereas discriminative models classify data by distinguishing between different classes
C Discriminative models are only used for text classification, while generative models are only used for image classification
D Generative models are trained on labeled data, while discriminative models can be trained on both labeled and unlabeled data

58 Which statement best describes the Amazon Personalize service?
A Derive and understand valuable insights from text within documents
B Deploy high-quality, natural-sounding human voices in dozens of languages
C Automatically convert speech to text and gain insights
D Elevate the customer experience with ML-powered personalization

59 A team needs to evaluate answer correctness and helpfulness where exact word overlap is too restrictive. Why might it use an LLM-as-a-judge instead of relying only on ROUGE?
A An evaluator LLM can apply semantic, rubric-based criteria instead of measuring only lexical overlap
B An evaluator LLM guarantees that every score is objective and error-free
C ROUGE can evaluate only images, while an evaluator LLM can evaluate only audio
D LLM-as-a-judge does not require evaluation instructions or representative data

60 In terms of AI systems and security, what is "prompt injection"?
A A method where an attacker manipulates input prompts to control AI output
B A technique to automatically generate security
C A vulnerability in AWS network infrastructure
D A strategy for encrypting data in transit

61 Which statement best describes the Strands Agents SDK?
A It is an open-source, model-first SDK for building agents with models, tools, and an agent loop
B It is an AWS compliance report repository
C It is a proprietary foundation model that can run only in Amazon SageMaker AI
D It is a managed relational database for storing embeddings

62 In the context of data governance for AI systems on AWS, what is the primary difference between data residency and data logging?

A Data residency refers to where data is physically stored, while data logging tracks data access and changes over time
B Data residency is concerned with data encryption, while data logging focuses on data transformation processes
C Data residency tracks user activities within an AI system, while data logging determines where data can be geographically stored
D Data residency involves monitoring real-time data usage, while data logging manages data lifecycle policies

63 How can you test and fine-tune prompts and parameters before using them in an application with Amazon Bedrock?

A By deploying the model in a live environment
B By using the Bedrock playground
C By using only predefined prompts
D By writing custom scripts from scratch

64 What is the primary difference between Reinforcement Learning from Human Feedback (RLHF) and Amazon Augmented AI (A2I)?

A RLHF requires no human involvement during the training process, while A2I automates the entire machine learning workflow without human review
B RLHF is used exclusively for natural language processing tasks, whereas A2I is used for image recognition and analysis tasks
C RLHF focuses on automatically generating data labels for training datasets, while A2I is used for unsupervised learning tasks
D RLHF is a technique used to train AI models using human feedback to refine their behavior, whereas A2I is a service that provides a human review of machine learning predictions to improve model accuracy and reliability

65 Which deployment model is the right fit for workloads that can tolerate cold starts in Amazon SageMaker?

A Asynchronous Inference
B Batch transform
C Real-time hosting services
D Serverless Inference

RESPONSE SHEET

Mark your answers here

Fill one bubble per question, or two where the question says Select two. Letters shown in outline only exist for that question — a question with four options has no E. Score against the key on the next page.

1 A B C D E	2 A B C D E ×2	3 A B C D E	4 A B C D E	5 A B C D E
6 A B C D E	7 A B C D E	8 A B C D E	9 A B C D E	10 A B C D E
11 A B C D E ×2	12 A B C D E	13 A B C D E ×2	14 A B C D E	15 A B C D E ×2
16 A B C D E	17 A B C D E	18 A B C D E	19 A B C D E	20 A B C D E
21 A B C D E ×2	22 A B C D E	23 A B C D E	24 A B C D E	25 A B C D E
26 A B C D E	27 A B C D E	28 A B C D E	29 A B C D E	30 A B C D E
31 A B C D E	32 A B C D E ×2	33 A B C D E	34 A B C D E ×2	35 A B C D E
36 A B C D E ×2	37 A B C D E	38 A B C D E	39 A B C D E	40 A B C D E
41 A B C D E ×2	42 A B C D E	43 A B C D E	44 A B C D E	45 A B C D E ×2
46 A B C D E	47 A B C D E ×2	48 A B C D E	49 A B C D E	50 A B C D E
51 A B C D E	52 A B C D E	53 A B C D E	54 A B C D E	55 A B C D E
56 A B C D E ×2	57 A B C D E	58 A B C D E	59 A B C D E	60 A B C D E
61 A B C D E	62 A B C D E	63 A B C D E	64 A B C D E	65 A B C D E

FL AGGED FOR REVIEW

Write the number of any question you guessed or want to revisit. These are the ones to read the explanation for first, even when you happened to get them right.

ANSWER KEY

Score and diagnose

Each row gives the correct letter or letters and the domain the question belongs to. Tally your domain scores below to see where to study next.

1 D	D5	2 A, C	D3	3 D	D4	4 C	D2	5 D	D3
6 D	D1	7 A	D3	8 D	D2	9 B	D4	10 B	D2
11 A, E	D3	12 A	D3	13 D, E	D4	14 C	D3	15 A, E	D1
16 A	D5	17 A	D1	18 C	D2	19 B	D3	20 B	D3
21 D, E	D2	22 C	D2	23 C	D3	24 C	D1	25 A	D2
26 D	D4	27 A	D2	28 B	D2	29 D	D5	30 D	D3
31 B	D1	32 A, B	D2	33 B	D5	34 B, D	D1	35 C	D1
36 C, E	D3	37 A	D2	38 B	D5	39 C	D4	40 A	D2
41 A, B	D3	42 C	D5	43 A	D3	44 C	D3	45 B, D	D4
46 D	D2	47 A, D	D3	48 A	D5	49 B	D1	50 B	D3
51 D	D1	52 C	D2	53 A	D4	54 D	D3	55 D	D1
56 A, B	D4	57 B	D2	58 D	D1	59 A	D3	60 A	D5
61 A	D2	62 A	D5	63 B	D3	64 D	D4	65 D	D1

DOMAIN		YOUR SCORE	OUT OF	TARGET
D1	Fundamentals of AI and ML		12	9
D2	Fundamentals of Generative AI		16	12
D3	Applications of Foundation Models		19	14
D4	Guidelines for Responsible AI		9	6
D5	Security, Compliance, and Governance		9	6
Total			65	47

AWS scores AIF-C01 on a 100–1000 scale with 700 to pass and does not publish a question-count threshold. The targets above use roughly 72 percent as a working proxy: clear it comfortably in every domain before you sit the real exam.

PA R T T H R E E

Answers explained

Every question, with the correct response, the reasoning behind it, and a primary AWS source you can read afterwards. Work through the ones you missed first, then skim the rest — the distractors are where most of the learning is.

D5 · SECURITY, COMPLIANCE, AND GOVERNANCE

1

How does AWS Identity and Access Management (IAM) help ensure security and compliance?
D Granular Access Control

IAM provides granular authentication and authorization through identities, roles, policies, permissions, and conditions.
REFERENCE What is IAM? https://docs.aws.amazon.com/IAM/latest/UserGuide/introduction.html

D3 · APPLICATIONS OF FOUNDATION MODELS

2

Which of the followings are best practices for creating effective prompts?
A + C Be clear and concise; Include relevant context

Effective prompts are clear and concise and include the context needed to perform the task. Always using negative prompts or high temperature is not a best practice.
REFERENCE Amazon Bedrock Prompt management https://docs.aws.amazon.com/bedrock/latest/userguide/prompt-management.html

D4 · GUIDELINES FOR RESPONSIBLE AI

3

How does SageMaker Ground Truth support the capabilities needed for Reinforcement Learning from Human Feedback (RLHF)?
D SageMaker Ground Truth enables the creation of high-quality labeled datasets by incorporating human feedback in the labeling process, which can be used to improve reinforcement learning models

Amazon SageMaker Ground Truth offers the most comprehensive set of human-in-the-loop capabilities for incorporating human feedback across the ML lifecycle to improve model accuracy and relevancy. You can complete various human-in-the-loop tasks, from data generation and annotation to reward model generation, model review, and customization through a self-service or AWS managed offering. SageMaker Ground Truth helps in creating high-quality labeled datasets by incorporating human feedback, which is crucial for training and refining reinforcement learning models.
REFERENCE AWS documentation https://aws.amazon.com/what-is/reinforcement-learning-from-human-feedback/

D2 · FUNDAMENTALS OF GENERATIVE AI

4

What is a primary characteristic of diffusion models in generative AI?
C They generate data by progressively adding noise to the input and then learning to reverse the noise process.

Diffusion models add noise during training and learn the reverse denoising process used to generate new samples.
REFERENCE AIF-C01 Domain 2 exam guide https://docs.aws.amazon.com/aws-certification/latest/ai-practitioner-

0

1/ai-practitioner-01-domain2.html

5 D 3 · APPLICATIONS OF FOUNDATION MODELS

Which metric primarily focuses on evaluating the similarity between a machine-generated summary and a human-written one by checking for matching sequences of words?

D ROUGE

ROUGE evaluates overlap between generated and reference text and is commonly used for summarization; BERTScore focuses more on semantic similarity.

REFERENCE AIF-C01 Domain 3 exam guide https://docs.aws.amazon.com/aws-certification/latest/ai-practitioner-01/ai-practitioner-01-domain3.html

6 D 1 · FUNDAMENTALS OF AI AND ML

Consider a scenario where a fully-managed AWS service needs to be used for automating the extraction of insights from legal briefs such as contracts and court records. What do you recommend?

D Amazon Comprehend

Amazon Comprehend is a natural language processing (NLP) service that uses machine learning to find meaning and insights in text. Natural Language Processing (NLP) is a way for computers to analyze, understand, and derive meaning from textual information in a smart and useful way. By utilizing NLP, you can extract important phrases, sentiments, syntax, key entities such as brand, date, location, person, etc., and the language of the text.

REFERENCE AWS documentation https://aws.amazon.com/comprehend/

7 D 3 · APPLICATIONS OF FOUNDATION MODELS

A manufacturing company aims to leverage Amazon Bedrock to create a generative AI application that automates the monitoring of inventory levels, sales data, and supply chain information. The application should also recommend optimal reorder points and quantities to enhance operational efficiency. What do you recommend?

A Agents for Amazon Bedrock

An agent is appropriate when an application must monitor information, reason over it, and orchestrate actions such as recommendations across systems.

REFERENCE How Amazon Bedrock Agents works https://docs.aws.amazon.com/bedrock/latest/userguide/agents-how.html

D 2 · F U N DA M E N TA L S O F G E N E R ATI V E A I

8

A telecom company wants to help its customer service agents provide better customer service by leveraging generative AI. Which of the following is the best fit for the given use case?

D Amazon Q in Connect

Amazon Connect is the contact center service from AWS. Amazon Q helps customer service agents provide better customer service. Amazon Q in Connect uses real-time conversation with the customer along with relevant company content to automatically recommend what to say or what actions an agent should take to better assist customers.

R E F E R E N C E AWS documentation https://aws.amazon.com/q/

D 4 · G U I D E L I N E S FO R R E S P O N S I B L E A I

9

Which Amazon SageMaker service will help you track the Machine Learning models that are hosted on endpoints for real-time inference ?

B Amazon SageMaker Model Dashboard

Amazon SageMaker Model Dashboard is a centralized portal, accessible from the SageMaker console, where you can view, search, and explore all of the models in your account. You can track which models are deployed for inference and if they are used in batch transform jobs or hosted on endpoints. If you set up monitors with Amazon SageMaker Model Monitor, you can also track the performance of your models as they make real-time predictions on live data.

R E F E R E N C E AWS documentation https://docs.aws.amazon.com/sagemaker/latest/dg/model-dashboard.html

D 2 · F U N DA M E N TA L S O F G E N E R ATI V E A I

10

What is the key difference between prompt engineering and context engineering?

B Prompt engineering focuses on how instructions are written, while context engineering designs the system that assembles the relevant information shown to the model

Prompt engineering primarily shapes the request and instructions. Context engineering is broader: it dynamically selects and assembles the information the model needs, which can include instructions, retrieved data, memory, tool results, and conversation state.

R E F E R E N C E Understanding context engineering for generative AI https://docs.aws.amazon.com/prescriptive-guidance/latest/gen-ai-lifecycle-operational-excellence/dev-experimenting-context.html

D 3 · A P P L I CATI O N S O F FO U N DATI O N M O D E L S

11

In the development of AI agents, which two factors are most critical for ensuring that the agent can effectively adapt to new and unpredictable environments?

A + E Enabling the agent to receive real-time feedback and adjust its actions accordingly; Incorporating reinforcement learning to allow the agent to learn from its environment over time

Real-time feedback plus reinforcement learning allows an agent to adjust behavior from experience instead of depending on a fixed rule set.

R E F E R E N C E How Amazon Bedrock Agents works https://docs.aws.amazon.com/bedrock/latest/userguide/agents-how.html

D 3 · A P P L I CATI O N S O F FO U N DATI O N M O D E L S

12

How does Amazon Bedrock use a teacher model during a model-distillation job?

A It uses teacher responses to create synthetic training data for fine-tuning the student model

Bedrock sends prepared prompts to the teacher and uses the resulting high-quality responses as synthetic data. That data is then used to fine-tune the selected student model for the target use case.

R E F E R E N C E Customize a model with distillation in Amazon Bedrock https://docs.aws.amazon.com/bedrock/latest/userguide/model-distillation.html

D 4 · G U I D E L I N E S FO R R E S P O N S I B L E A I

13

Which AWS services are specifically designed to aid in monitoring machine learning models and incorporating human review processes?

D + E Amazon Augmented AI (Amazon A2I); Amazon SageMaker Model Monitor

Amazon SageMaker Model Monitor is a service that continuously monitors the quality of machine learning models in production and helps detect data drift, model quality issues, and anomalies. It ensures that models perform as expected and alerts users to issues that might require human intervention. Amazon Augmented AI (A2I) is a service that helps implement human review workflows for machine learning predictions.

R E F E R E N C E AWS documentation https://docs.aws.amazon.com/sagemaker/latest/dg/a2i-use-augmented-ai-a2i-human-review-loops.html

D 3 · A P P L I CATI O N S O F FO U N DATI O N M O D E L S

14

Which of the following best describes an action group in the context of setting up an AI agent?

C A collection of tasks that the agent can perform to help the user

An action group defines actions an agent can perform, typically through an API or function schema and optional Lambda integration.

R E F E R E N C E How Amazon Bedrock Agents works https://docs.aws.amazon.com/bedrock/latest/userguide/agents-how.html

D 1 · F U N DA M E N TA L S O F A I A N D M L

15

Which of the following are examples of unsupervised learning?
A + E Clustering; Dimensionality reduction

Unsupervised learning algorithms train on unlabeled data. They scan through new data and establish meaningful connections between the unknown input and predetermined outputs. For instance, unsupervised learning algorithms could group news articles from different news sites into common categories like sports and crime.
R E F E R E N C E AWS documentation https://aws.amazon.com/compare/the-difference-between-machine-learning-supervised-and-unsupervised/

D 5 · S E C U R IT Y, CO M P L I A N C E , A N D G OV E R N A N C E

16

A team wants to investigate an agent's production latency, errors, tool calls, and execution path. Which approach should it use?
A Enable AgentCore observability and use Amazon CloudWatch metrics, traces, and spans

AgentCore provides service metrics in CloudWatch and can produce traces, spans, and logs for supported resources. CloudWatch Transaction Search must be enabled to view the generated traces and spans.
R E F E R E N C E Configure AgentCore observability https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/observability-configure.html

D 1 · F U N DA M E N TA L S O F A I A N D M L

17

What is a key difference between labeled data and unlabeled data in the context of machine learning?
A Labeled data is annotated with output labels that provide specific information about each data point and is used for supervised learning, whereas, unlabeled data lacks such annotations and is used for unsupervised learning

Labeled data is data that comes with predefined labels or annotations. Each data point has an associated label that provides information about the data. This type of data is crucial for supervised learning, where the model learns to predict the output from the input data.
R E F E R E N C E AWS documentation https://aws.amazon.com/what-is/data-labeling/

18
D 2 · F U N DA M E N TA L S O F G E N E R ATI V E A I

What is a key advantage of using foundational models in generative AI applications?
C They are pre-trained on vast datasets, allowing them to be fine-tuned for specific tasks with minimal additional training

Foundation models are pretrained on broad datasets and can be adapted to downstream tasks, reducing the data and effort needed compared with training from scratch.
R E F E R E N C E AIF-C01 Domain 2 exam guide https://docs.aws.amazon.com/aws-certification/latest/ai-practitioner-0
1/ai-practitioner-01-domain2.html

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D 3 · A P P L I CATI O N S O F FO U N DATI O N M O D E L S

Which AgentCore Memory feature most directly helps an agent personalize later conversations based on a preference stated in an earlier session?
B Long-term memory that extracts and stores user preferences across sessions

Long-term memory persists extracted insights such as preferences, facts, and summaries across sessions. This information can be retrieved later and placed into the agent's context to support personalized responses.
R E F E R E N C E Add memory to an AgentCore agent https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/mem
ory.html

20
D 3 · A P P L I CATI O N S O F FO U N DATI O N M O D E L S

A company is using the Amazon Titan Text model with Amazon Bedrock. In which of the following scenarios is the model most likely to hallucinate?
B When temperature is set to 1

Temperature is a value between 0 and 1, and it regulates the creativity of the model's responses. Use a lower temperature if you want more deterministic responses. Use a higher temperature if you want creative or different responses for the same prompt on Amazon Bedrock and this is how you might see hallucination responses.
R E F E R E N C E AWS documentation https://docs.aws.amazon.com/bedrock/latest/userguide/inference-parameters.html

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D 2 · F U N DA M E N TA L S O F G E N E R ATI V E A I

Which of the following are the features of Amazon SageMaker JumpStart?

D + E Pre-trained models are fully customizable for your use case with your data; You can evaluate, compare, and select Foundation Models quickly based on pre-defined quality and responsibility metrics

Amazon SageMaker JumpStart is a machine learning (ML) hub that can help you accelerate your ML journey. With SageMaker JumpStart, you can evaluate, compare, and select FMs quickly based on pre-defined quality and responsibility metrics to perform tasks like article summarization and image generation. Pretrained models are fully customizable for your use case with your data, and you can easily deploy them into production with the user interface or SDK.

R E F E R E N C E AWS documentation https://aws.amazon.com/sagemaker/jumpstart/

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D 2 · F U N DA M E N TA L S O F G E N E R ATI V E A I

How does Amazon Bedrock evaluate the pricing for image models?

C Based on the number of images generated

For supported image-generation models, Bedrock on-demand pricing is based on images generated, with rates varying by model and image settings.

R E F E R E N C E What is Amazon Bedrock? https://docs.aws.amazon.com/bedrock/latest/userguide/what-is-bedrock.html

23

D 3 · A P P L I CATI O N S O F FO U N DATI O N M O D E L S

An insurance company is transitioning to AWS Cloud and wants to use Amazon Bedrock for product recommendations. The company wants to supplement organization-specific information to the underlying Foundation Model (FM). Which of the following represents the best-fit solution for the given use case?

C Use Knowledge Bases for Amazon Bedrock to supplement contextual information from the company's private data to the FM using Retrieval Augmented Generation (RAG)

With the comprehensive capabilities of Amazon Bedrock, you can experiment with a variety of top FMs, customize them privately with your data using techniques such as fine-tuning and retrieval-augmented generation (RAG), and create managed agents that execute complex business tasks—from booking travel and processing insurance claims to creating ad campaigns and managing inventory—all without writing any code. Using Knowledge Bases for Amazon Bedrock, you can provide foundation models with contextual information from your company's private data for Retrieval Augmented Generation (RAG), enhancing response relevance and accuracy.

R E F E R E N C E AWS documentation https://docs.aws.amazon.com/bedrock/latest/userguide/knowledge-base.html

D 1 · F U N DA M E N TA L S O F A I A N D M L

24

A company needs a solution that can convert text into human speech so that it can offer audio courses in multiple languages. Which AWS service is the best fit for this use case ?

C Amazon Polly

Amazon Polly uses deep learning technologies to synthesize natural-sounding human speech, so you can convert articles to speech. With dozens of lifelike voices across a broad set of languages, use Amazon Polly to build speech-activated applications. Amazon Polly is a service that turns text into lifelike speech.

R E F E R E N C E AWS documentation https://aws.amazon.com/polly/

D 2 · F U N DA M E N TA L S O F G E N E R ATI V E A I

25

What are the applications of large language models (LLMs)?

A LLMs are used for generating human-like text, translating languages, summarizing text, and answering questions based on large datasets

LLMs are designed for language tasks such as generation, translation, summarization, and question answering. Video, 3D, and speech synthesis require other model modalities or services.

R E F E R E N C E AIF-C01 Domain 2 exam guide https://docs.aws.amazon.com/aws-certification/latest/ai-practitioner-
0
1/ai-practitioner-01-domain2.html

D 4 · G U I D E L I N E S FO R R E S P O N S I B L E A I

26

What is the bias versus variance trade-off in machine learning?

D The bias versus variance trade-off refers to the challenge of balancing the error due to the model's complexity (variance) and the error due to incorrect assumptions in the model (bias), where high bias can cause underfitting and high variance can cause overfitting

The bias versus variance trade-off in machine learning is about finding a balance between bias (error due to overly simplistic assumptions in the model, leading to underfitting) and variance (error due to the model being too sensitive to small fluctuations in the training data, leading to overfitting). The goal is to achieve a model that generalizes well to new data.

R E F E R E N C E AWS documentation https://docs.aws.amazon.com/wellarchitected/latest/machine-learning-lens/mlper-09.html

D 2 · F U N DA M E N TA L S O F G E N E R ATI V E A I

27

A team repeatedly explains its architecture, naming conventions, and preferred libraries in every Kiro conversation. Which capability should it configure?

A Steering files in .kiro/steering/

Kiro steering provides persistent project knowledge such as product goals, technology choices, project structure, coding patterns, and standards. Workspace steering files allow this guidance to be shared with the project instead of repeated in each chat.

R E F E R E N C E Kiro steering https://kiro.dev/docs/steering/

D 2 · F U N DA M E N TA L S O F G E N E R ATI V E A I

28

What is a key difference between Amazon Bedrock and Amazon SageMaker JumpStart?

B Amazon Bedrock provides foundational models for generative AI applications, whereas Amazon SageMaker JumpStart offers pre-built solutions and one-click deployment for various machine learning models

Amazon Bedrock is the easiest way to build and scale generative AI applications with foundation models. Amazon Bedrock is a fully managed service that offers a choice of high-performing foundation models (FMs) from leading AI companies like AI21 Labs, Anthropic, Cohere, Meta, Mistral AI, Stability AI, and Amazon through a single API, along with a broad set of capabilities you need to build generative AI applications with security, privacy, and responsible AI. Amazon SageMaker JumpStart is a machine learning (ML) hub that can help you accelerate your ML journey.

R E F E R E N C E AWS documentation https://aws.amazon.com/bedrock/

D 5 · S E C U R IT Y, CO M P L I A N C E , A N D G OV E R N A N C E

29

In the context of the AWS Shared Responsibility Model, which statement best describes the security responsibilities of both AWS and the customer when using Amazon Bedrock for generative AI applications?

D AWS is responsible for securing the infrastructure that runs Amazon Bedrock, while the customer is responsible for securing their data and managing access controls

According to the AWS Shared Responsibility Model, AWS manages the security of the cloud, which includes protecting the infrastructure that runs Amazon Bedrock services, such as hardware, software, networking, and facilities. Customers, on the other hand, are responsible for security in the cloud, which includes managing their data, configuring access controls, and ensuring that AI models and applications are securely implemented. For Bedrock, customers are responsible for data management, access controls, and setting up guardrails.

R E F E R E N C E AWS documentation https://aws.amazon.com/compliance/shared-responsibility-model/

D 3 · A P P L I CATI O N S O F FO U N DATI O N M O D E L S

30

How does the inference parameter Response length influence the model response for Amazon Bedrock?

D Specifies the minimum or maximum number of tokens to return in the generated response.

Response length represents the minimum or maximum number of tokens to return in the generated response.
R E F E R E N C E AWS documentation https://docs.aws.amazon.com/bedrock/latest/userguide/inference-parameters.html

D 1 · F U N DA M E N TA L S O F A I A N D M L

31

A company needs to extract handwritten words and letters from scanned documents. Which ML-powered AWS service is the right fit for this requirement?

B Amazon Textract

Amazon Textract is a machine learning (ML) service that automatically extracts text, handwriting, layout elements, and data from scanned documents. It goes beyond simple optical character recognition (OCR) to identify, understand, and extract data from forms and tables. All extracted data is returned with bounding box coordinates—polygon frames that encompass each piece of identified data, such as a word, a line, a table, or individual cells within a table.
R E F E R E N C E AWS documentation https://aws.amazon.com/textract/faqs/

D 2 · F U N DA M E N TA L S O F G E N E R ATI V E A I

32

Which of the following options summarize the differences between Amazon Q and Amazon Bedrock?

A + B Amazon Q is a generative AI–powered assistant that allows you to create pre-packaged generative AI applications, whereas, Amazon Bedrock provides an environment to build and scale generative AI applications using a Foundation Model (FM); With Amazon Bedrock, you can choose the underlying Foundation Model. However, Amazon Q does not allow you to choose the underlying Foundation Model

Amazon Q is a generative AI-powered assistant for accelerating software development and leveraging companies' internal data. Amazon Q generates code, tests, and debugs. It has multistep planning and reasoning capabilities that can transform and implement new code generated from developer requests.
R E F E R E N C E AWS documentation https://aws.amazon.com/q/

33

D 5 · SECURITY, COMPLIANCE, AND GOVERNANCE

In the context of security and privacy for AI systems on AWS, what is the primary difference between threat detection and vulnerability management?

B Threat detection involves real-time monitoring and identification of active threats, whereas vulnerability management is about identifying, assessing, and mitigating security weaknesses

Threat detection services, such as Amazon GuardDuty, continuously monitor the environment to identify active threats and malicious activities in real time, providing alerts and insights to respond promptly. Vulnerability management, on the other hand, focuses on proactively identifying, assessing, and mitigating security vulnerabilities before they can be exploited. This involves regular scanning, patch management, and implementing security best practices.

REFERENCE AWS documentation https://aws.amazon.com/ai/generative-ai/security/

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D 1 · FUNDAMENTALS OF AI AND ML

Which of the following is correct regarding the training set, validation set, and test set used in the context of machine learning?

B + D Validation sets are optional; Test set is used to determine how well the model generalizes

Data used for ML is typically split into the following datasets: The training set is used to train the model, the validation set is used for tuning hyperparameters and selecting the best model during the training process, and the test set is used for evaluating the final performance of the model on unseen data. The validation set introduces new data to the trained model. You can use a validation set to periodically measure model performance as training is happening and also tune any hyperparameters of the model.

REFERENCE AWS documentation https://aws.amazon.com/blogs/machine-learning/create-train-test-and-validation-splits-on-your-data-for-machine-learning-with-amazon-sagemaker-data-wrangler/

35

D 1 · FUNDAMENTALS OF AI AND ML

Which AWS service offers pre-trained and customizable computer vision (CV) capabilities?

C Amazon Rekognition

Amazon Rekognition is a cloud-based image and video analysis service that makes it easy to add advanced computer vision capabilities to your applications. The service is powered by proven deep learning technology and it requires no machine learning expertise to use. Amazon Rekognition includes a simple, easy-to-use API that can quickly analyze any image or video file that’s stored in Amazon S3.

REFERENCE AWS documentation https://docs.aws.amazon.com/rekognition/latest/dg/what-is.html

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D 3 · A P P L I CATI O N S O F FO U N DATI O N M O D E L S

Which of the following are key advantages of using "few-shot" prompting in prompt engineering?
C + E It helps the model understand the task by providing examples.; It allows for more accurate results with complex tasks by guiding the model.

Few-shot examples clarify the intended task and can improve accuracy on complex or ambiguous inputs; they do not eliminate randomness or examples.
R E F E R E N C E Amazon Bedrock Prompt management https://docs.aws.amazon.com/bedrock/latest/userguide/prompt-management.html

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D 2 · F U N DA M E N TA L S O F G E N E R ATI V E A I

What sequence best represents the core Strands agent loop?
A Invoke the model, execute a requested tool, return the tool result to the model, and repeat until a final response

In the Strands agent loop, the model decides whether it needs a tool. The execution system validates and runs the requested tool, adds its result to the conversation, and invokes the model again until the model produces a final answer or another stop condition occurs.
R E F E R E N C E Strands agent loop https://strandsagents.com/docs/user-guide/concepts/agents/agent-loop/

38
D 5 · S E C U R IT Y, CO M P L I A N C E , A N D G OV E R N A N C E

Which of the following is correct regarding data security and compliance aspects of Amazon Bedrock?
B Your data is not used to improve the base Foundation Models (FMs) and it is not shared with any model providers

Amazon Bedrock is a fully managed service that makes high-performing foundation models (FMs) from leading AI startups and Amazon available for your use through a unified API. Using Amazon Bedrock, you can easily experiment with and evaluate top foundation models for your use cases, privately customize them with your data using techniques such as fine-tuning and Retrieval Augmented Generation (RAG), and build agents that execute tasks using your enterprise systems and data sources. With Amazon Bedrock, your content is not used to improve the base models and is not shared with any model providers.
R E F E R E N C E AWS documentation https://aws.amazon.com/bedrock/faqs/

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D 4 · G U I D E L I N E S FO R R E S P O N S I B L E A I

Which of the following scenarios best illustrates algorithmic bias?
C A hiring algorithm consistently prefers candidates from a particular gender, even though the candidates' qualifications are similar across genders

Biases are imbalances in data or disparities in the performance of a model across different groups. Bias may also be introduced by the ML algorithm itself—even with a well-balanced training dataset, the outcomes might favor certain subsets of the data as compared to others. This scenario illustrates algorithmic bias, where the hiring algorithm systematically favors candidates of a particular gender, indicating that there may be a bias in the training data or in the algorithm's design that leads to unequal treatment based on gender.
R E F E R E N C E AWS documentation https://aws.amazon.com/blogs/machine-learning/learn-how-amazon-sagemaker-clarify-helps-detect-bias/

40

D 2 · F U N DA M E N TA L S O F G E N E R ATI V E A I

Which Kiro hook action is the better choice for running a fixed linter command whenever a relevant event occurs?
A Shell Command, because the action is deterministic

Kiro hooks support agent-prompt and shell-command actions. A shell command is appropriate for a deterministic operation such as invoking a linter, while an agent prompt is useful when the response requires context-sensitive natural-language reasoning.
R E F E R E N C E Kiro hook actions https://kiro.dev/docs/hooks/actions/

41

D 3 · A P P L I CATI O N S O F FO U N DATI O N M O D E L S

Which of the following generative AI techniques are used in Amazon Q Business web application workflow?
A + B Large Language Model (LLM); Retrieval-Augmented Generation (RAG)

Large language models (LLMs) are a class of Foundation Models (FMs). For example, OpenAI's generative pre-trained transformer (GPT) models are LLMs. LLMs are specifically focused on language-based tasks such as such as summarization, text generation, classification, open-ended conversation, and information extraction.
R E F E R E N C E AWS documentation https://docs.aws.amazon.com/amazonq/latest/qbusiness-ug/how-it-works.html

D 5 · S E C U R IT Y, CO M P L I A N C E , A N D G OV E R N A N C E

42

Which feature of Amazon Macie enables organizations to automatically identify and safeguard sensitive data within their Amazon S3 buckets?

C Automated Data Discovery

Amazon Macie automated sensitive-data discovery continually evaluates S3 objects to identify sensitive data and produce findings for investigation and protection workflows.

R E F E R E N C E What is Amazon Macie? https://docs.aws.amazon.com/macie/latest/user/what-is-macie.html

D 3 · A P P L I CATI O N S O F FO U N DATI O N M O D E L S

43

What is the role of vector embeddings in Retrieval-Augmented Generation (RAG)?

A They represent words in a vector space to understand semantics

Embeddings encode semantic meaning as vectors so a RAG system can retrieve chunks that are similar in meaning to the user query.

R E F E R E N C E Knowledge Bases for Amazon Bedrock https://docs.aws.amazon.com/bedrock/latest/userguide/knowledge-base.html

D 3 · A P P L I CATI O N S O F FO U N DATI O N M O D E L S

44

Which of the following is correct regarding the techniques used to improve the performance of a Foundation Model (FM)?

C Fine-tuning changes the weights of the FM whereas Retrieval-augmented generation (RAG) does not change the weights of the FM

Retrieval-Augmented Generation (RAG) is the process of optimizing the output of a large language model, so it references an authoritative knowledge base outside of its training data sources before generating a response. Large Language Models (LLMs) are trained on vast volumes of data and use billions of parameters to generate original output for tasks like answering questions, translating languages, and completing sentences. RAG extends the already powerful capabilities of LLMs to specific domains or an organization's internal knowledge base, all without the need to retrain the model.

R E F E R E N C E AWS documentation https://aws.amazon.com/blogs/machine-learning/best-practices-to-build-generative-ai-applications-on-aws/

45

D 4 · G U I D E L I N E S FO R R E S P O N S I B L E A I

Which AWS services/tools can be used to implement Responsible AI practices?
B + D Amazon SageMaker Model Monitor; Amazon SageMaker Clarify

Amazon SageMaker Clarify is a service provided by AWS (Amazon Web Services) that helps developers detect biases and explain the predictions made by machine learning models. It is part of the Amazon SageMaker suite of machine learning tools and focuses on enhancing transparency, fairness, and explainability in machine learning workflows. Amazon SageMaker Model Monitor is a service within the Amazon SageMaker suite that helps developers continuously monitor machine learning models deployed in production.
R E F E R E N C E AWS documentation https://aws.amazon.com/machine-learning/responsible-ai/resources/

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D 2 · F U N DA M E N TA L S O F G E N E R ATI V E A I

Which AWS service/feature offers a choice of high-performing Foundation Models (FMs) and the ability to privately customize the FMs with your data?
D Amazon Bedrock

Amazon Bedrock is a fully managed service that offers a choice of high-performing FMs and a broad set of capabilities. You can easily experiment with various top FMs, privately customize them with your data, and create managed agents that execute complex business tasks. Amazon Bedrock is the easiest way to build and scale generative AI applications with foundation models.
R E F E R E N C E AWS documentation https://aws.amazon.com/what-is/generative-ai/

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D 3 · A P P L I CATI O N S O F FO U N DATI O N M O D E L S

Which of the following parameters can influence the creativity of a language model's output?
A + D Temperature; Top-p

Temperature and Top-P both influence output diversity: temperature reshapes token probabilities, while Top-P limits sampling to a cumulative-probability set.
R E F E R E N C E Amazon Bedrock inference parameters https://docs.aws.amazon.com/bedrock/latest/userguide/inference-parameters.html

D 5 · S E C U R IT Y, CO M P L I A N C E , A N D G OV E R N A N C E

48

When configuring an AgentCore Gateway, what is the distinction between inbound and outbound authorization?

A Inbound authorization controls access to the gateway; outbound authorization controls how the gateway accesses target resources

Inbound authorization determines which clients may call the Gateway endpoint. Outbound authorization supplies the credentials and controls needed when Gateway invokes a configured backend target on behalf of the caller.

R E F E R E N C E Set up an AgentCore gateway https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/gateway-building.html

D 1 · F U N DA M E N TA L S O F A I A N D M L

49

Which of the following best represents the capabilities of Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs)?

B While CNNs are used for single image analysis, RNNs are used for video analysis

CNNs are a type of deep learning model particularly well-suited for processing grid-like data, such as images. They are designed to automatically and adaptively learn spatial hierarchies of features from input images. RNNs are designed to handle sequential data, where the order of the data points matters.

R E F E R E N C E AWS documentation https://aws.amazon.com/what-is/computer-vision/

D 3 · A P P L I CATI O N S O F FO U N DATI O N M O D E L S

50

A company wants to improve the performance of a Foundation Model (FM) being used in Amazon Bedrock. Which of the following lists the underlying techniques in the increasing order of complexity for implementing a solution?

B Prompt engineering, Retrieval Augmented Generation (RAG), Fine-tuning

Prompt engineering is the practice of carefully designing prompts to efficiently tap into the capabilities of FMs. It involves the use of prompts, which are short pieces of text that guide the model to generate more accurate and relevant responses. With prompt engineering, you can improve the performance of FMs and make them more effective for a variety of applications.

R E F E R E N C E AWS documentation https://aws.amazon.com/blogs/machine-learning/best-practices-to-build-generative-ai-applications-on-aws/

D 1 · F U N DA M E N TA L S O F A I A N D M L

51

What is the difference between computer vision and image processing?

D Image processing focuses on enhancing and manipulating images for visual quality, whereas computer vision involves interpreting and understanding the content of images to make decisions

Image processing is primarily concerned with the techniques used to enhance and manipulate images, such as filtering, noise reduction, and image transformation. Computer vision, on the other hand, focuses on interpreting and understanding the content of images to make decisions, such as object detection, facial recognition, and scene understanding. Computer vision often uses machine learning algorithms to achieve these tasks.

R E F E R E N C E AWS documentation https://aws.amazon.com/what-is/computer-vision/

D 2 · F U N DA M E N TA L S O F G E N E R ATI V E A I

52

Which of the following aptly summarizes the capabilities of Foundation Models?

C Foundation models can perform a wide range of tasks across different domains by leveraging their extensive pre-training on large datasets

Foundation models are a form of generative artificial intelligence (generative AI). They generate output from one or more inputs (prompts) in the form of human language instructions. In general, an FM uses learned patterns and relationships to predict the next item in a sequence.

R E F E R E N C E AWS documentation https://aws.amazon.com/what-is/foundation-models/

D 4 · G U I D E L I N E S FO R R E S P O N S I B L E A I

53

Which Amazon SageMaker service/tool offers a centralized view of all the Machine Learning models created in your AWS account?

A Amazon SageMaker Model Dashboard

SageMaker Model Dashboard provides a centralized view of models and integrates information from model cards, endpoints, and monitoring.

R E F E R E N C E Amazon SageMaker Model Dashboard https://docs.aws.amazon.com/sagemaker/latest/dg/model-dashboard.
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tml

D 3 · A P P L I CATI O N S O F FO U N DATI O N M O D E L S

54

The marketing department at a media company wants to leverage Amazon Bedrock for making creative scripts for an upcoming ad campaign. What do you recommend?

D Use higher Temperature to get more creative responses for the same prompt on Amazon Bedrock

Temperature is a value between 0 and 1, and it regulates the creativity of the model's responses. Use a lower temperature if you want more deterministic responses, and use a higher temperature if you want creative or different responses for the same prompt on Amazon Bedrock.
R E F E R E N C E AWS documentation https://docs.aws.amazon.com/bedrock/latest/userguide/inference-parameters.html

D 1 · F U N DA M E N TA L S O F A I A N D M L

55

A developer is working on an AI application for predicting customer churn. The developer is collaborating with a research team to ensure the best model is selected for the application. The application needs to accurately identify customers who are likely to leave the service within the next six months. What should the developer ask the research team to do in order to ensure that the best model is selected for the AI application?

D Define the use case of the application narrowly

A narrowly defined use case provides clear and specific requirements for the application, helping the research team understand exactly what the model needs to accomplish. This clarity is crucial for selecting the most appropriate model that fits the specific needs and constraints of the application.
R E F E R E N C E AWS documentation https://aws.amazon.com/what-is/machine-learning/

D 4 · G U I D E L I N E S FO R R E S P O N S I B L E A I

56

Which practices improve the reliability of an LLM-as-a-judge evaluation?

A + B Define clear evaluation criteria and rating scales; Use representative evaluation examples and validate results against human judgment

A judge model is still an AI model and can be biased or inconsistent. Clear rubrics, representative datasets, reference answers where appropriate, calibration, and comparison with human evaluation make its scores more meaningful and auditable.
R E F E R E N C E Define Amazon Bedrock evaluation methods https://docs.aws.amazon.com/bedrock/latest/userguide/advance
d-prompt-optimization-evaluation.html

D 2 · F U N DA M E N TA L S O F G E N E R ATI V E A I

57

What is the primary distinction between discriminative models and generative models in the context of generative AI?

B Generative models focus on generating new data from learned patterns, whereas discriminative models classify data by distinguishing between different classes

Generative models learn the underlying patterns of data to create new, similar data, while discriminative models learn to distinguish between different classes of data. Generative models, such as GPT-3, can generate new content, whereas discriminative models are used for classification tasks. The former focuses on understanding and replicating the data distribution, while the latter focuses on decision boundaries to classify inputs.

R E F E R E N C E AWS documentation https://aws.amazon.com/what-is/generative-ai/

D 1 · F U N DA M E N TA L S O F A I A N D M L

58

Which statement best describes the Amazon Personalize service?

D Elevate the customer experience with ML-powered personalization

Amazon Personalize is a fully managed machine learning (ML) service that uses your data to generate product and content recommendations for your users. You provide data about your end-users (e.g., age, location, device type), items in your catalog (e.g., genre, price), and interactions between users and items (e.g., clicks, purchases). Personalize uses this data to train custom, private models that generate recommendations that can be surfaced via an API.

R E F E R E N C E AWS documentation https://aws.amazon.com/personalize/faqs/

D 3 · A P P L I CATI O N S O F FO U N DATI O N M O D E L S

59

A team needs to evaluate answer correctness and helpfulness where exact word overlap is too restrictive. Why might it use an LLM-as-a-judge instead of relying only on ROUGE?

A An evaluator LLM can apply semantic, rubric-based criteria instead of measuring only lexical overlap

Metrics such as ROUGE measure overlap with reference text. An LLM judge can apply criteria such as correctness, completeness, relevance, or style to semantically equivalent answers, but its judgments still require validation because the evaluator can introduce bias or inconsistency.

R E F E R E N C E Amazon Bedrock judge-based evaluation metrics https://docs.aws.amazon.com/bedrock/latest/userguide/model-evaluation-metrics.html

D 5 · S E C U R IT Y, CO M P L I A N C E , A N D G OV E R N A N C E
60
In terms of AI systems and security, what is "prompt injection"?
A A method where an attacker manipulates input prompts to control AI output

Prompt injection is an attempt to manipulate model instructions through untrusted input so that the model ignores intended rules or performs unintended behavior.
R E F E R E N C E Common prompt attacks and mitigations https://docs.aws.amazon.com/prescriptive-guidance/latest/ll
m-pr
ompt-engineering-best-practices/common-attacks.html

D 2 · F U N DA M E N TA L S O F G E N E R ATI V E A I
61
Which statement best describes the Strands Agents SDK?
A It is an open-source, model-first SDK for building agents with models, tools, and an agent loop

Strands Agents is an open-source SDK initially released by AWS. Its model-first approach places the foundation model at the center of reasoning while tools extend the actions and information available to the agent.
R E F E R E N C E Strands Agents in AWS Prescriptive Guidance https://docs.aws.amazon.com/prescriptive-guidance/lat
est/ag
entic-ai-frameworks/strands-agents.html

D 5 · S E C U R IT Y, CO M P L I A N C E , A N D G OV E R N A N C E
62
In the context of data governance for AI systems on AWS, what is the primary difference between data residency and data logging?
A Data residency refers to where data is physically stored, while data logging tracks data access and changes over time

Data residency concerns the geographic location in which data is stored or processed; data logging records access, events, and changes over time.
R E F E R E N C E AIF-C01 Domain 5 exam guide https://docs.aws.amazon.com/aws-certification/latest/ai-practitioner-
0
1/ai-practitioner-01-domain5.html

D 3 · A P P L I CATI O N S O F FO U N DATI O N M O D E L S
63
How can you test and fine-tune prompts and parameters before using them in an application with Amazon Bedrock?
B By using the Bedrock playground

The Bedrock playground lets developers experiment with supported models, prompts, and inference parameters before application integration.
R E F E R E N C E Amazon Bedrock Prompt management https://docs.aws.amazon.com/bedrock/latest/userguide/prompt-mana
gement.html

D 4 · G U I D E L I N E S FO R R E S P O N S I B L E A I

64

What is the primary difference between Reinforcement Learning from Human Feedback (RLHF) and Amazon Augmented AI (A2I)?

D RLHF is a technique used to train AI models using human feedback to refine their behavior, whereas A2I is a service that provides a human review of machine learning predictions to improve model accuracy and reliability

Reinforcement learning from human feedback (RLHF) is a machine learning (ML) technique that uses human feedback to optimize ML models to self-learn more efficiently. Reinforcement learning (RL) techniques train software to make decisions that maximize rewards, making their outcomes more accurate. RLHF incorporates human feedback in the rewards function, so the ML model can perform tasks more aligned with human goals, wants, and needs.

R E F E R E N C E AWS documentation https://aws.amazon.com/what-is/reinforcement-learning-from-human-feedback/

D 1 · F U N DA M E N TA L S O F A I A N D M L

65

Which deployment model is the right fit for workloads that can tolerate cold starts in Amazon SageMaker?

D Serverless Inference

On-demand Serverless Inference is ideal for workloads that have idle periods between traffic spurts and can tolerate cold starts. Amazon SageMaker Serverless Inference is a purpose-built inference option that enables you to deploy and scale ML models without configuring or managing any of the underlying infrastructure. Serverless endpoints automatically launch compute resources and scale them in and out depending on traffic, eliminating the need to choose instance types or manage scaling policies.

R E F E R E N C E AWS documentation https://docs.aws.amazon.com/sagemaker/latest/dg/serverless-endpoints.html